# PENGZHEN LI

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## SUMMARY

Economics Ph.D. candidate (expected May 2027) combining causal inference, structural estimation, and large-scale optimization. Built a stochastic supply-chain optimization model spanning 1,374 U.S. counties, estimated a strategic game among Brazil's 5,570 mayors, and ran a triple-difference design on the universe of Chinese customs records (2000–2014) for my job market paper. Four peer-reviewed publications (one with 1,000+ citations) and six working papers. Python, R, Stata, SQL, GAMS. Authorized to work in the U.S. without sponsorship.

## EDUCATION

**University of Tennessee, Knoxville** Jan 2021 – May 2027 (expected)
Ph.D. in Economics — fields: international trade, applied econometrics, environmental and resource economics

**Shandong University, Jinan** Sep 2016 – Jun 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (1,000+ citations)

**Tongji University, Shanghai** Sep 2011 – Jun 2015
B.A. in Economics

## SKILLS

**Programming / software:** Python (pandas, numpy, statsmodels, scikit-learn), R, Stata, SQL, GAMS, MATLAB, Git, LaTeX; ArcGIS.
**Methods:** causal inference (difference-in-differences, event study, IV, RDD, synthetic control, fixed effects); structural estimation; panel and time-series econometrics (GMM, threshold models, VAR/VECM); forecasting; mixed-integer and stochastic optimization; game theory; machine learning; text-as-data.
**Credentials / languages:** CFA Level II; English (fluent), Mandarin (native).

## RESEARCH EXPERIENCE

**Doctoral Researcher**, Department of Economics, University of Tennessee 2023 – present

- *Market Access and the Quality of Chinese Exports* (job market paper): triple-difference design on the universe of Chinese customs records (2000–2014) around the 2002 removal of U.S. tariff uncertainty; measured export quality fell about 11% for heavily exposed products, and 83% of the estimate remains inside continuing firm–product–destination cells, so it is not explained by entry and exit alone. Showed that the price, quantity and quality estimates such designs report contain only two independent facts, and built a model in which wider market reach lowers the return to product quality.
- *Trade Restrictiveness Indices with Discriminatory Tariffs* (with J. Lake): extended the welfare-based tariff index to preferential (non-MFN) tariffs with large-country effects; separated where-to-buy from what-to-buy distortions and computed the index for 123 importers (2000–2020) from BACI/WITS data; in 2018 preferences cut the median simple average tariff by 16% but the index by only 5%. Presented at the Midwest Trade Meetings, 2025.
- *Section 45Z clean-fuel production credit* (two papers, with T. E. Yu): built a two-stage stochastic mixed-integer program siting a sustainable-aviation-fuel supply chain across 1,374 Southeast U.S.

counties; price risk justifies at most \$1.04/gal against the statutory \$1.30–\$1.45, and a one-sided price floor delivers the same output at 69% of the flat credit's expected cost. A companion Stackelberg model splits the remaining subsidy gap into risk-aversion and project-finance premia.

- *A Mean-Field Game of Mayoral Endorsements* (with M. Van Essen, L. Lima; 2024 Holly Award): recast a strategic game among Brazil's 5,570 mayors—too large to enumerate—as a low-dimensional fixed point estimated by least squares in milliseconds; a close-race difference-in-differences design finds discretionary transfers to co-partisan mayors 10.6% higher.
- *A Weight-Free Benchmark for Multilateral Tariff Bargaining* (with G. Schaur): replaced estimated Nash bargaining weights with a Shapley-value benchmark in a calibrated general-equilibrium trade model; the bargained outcome falls short of the full-schedule optimum by 79.5% of the group's surplus, mostly from limited agenda coverage.

**Graduate Research Assistant**, University of Tennessee Jan 2021 – Aug 2023

USDA NIFA and FAA ASCENT funded projects on sustainable-aviation-fuel supply chains.

- Built mixed-integer optimization models (GAMS) to design cost- and profit-optimal forest-residue sustainable-aviation-fuel supply chains across the Southeast U.S.; published in *Biomass and Bioenergy* (2026) and *Transportation Research Record* (2024).
- Ran facility-siting and regional impact analysis with IMPLAN and efficiency analysis with DEA; delivered scenario and sensitivity results (EMP, supervised ML, @RISK) to an interdisciplinary engineering–economics team and federal sponsors.
- Engineered reproducible ArcGIS/R/Python pipelines for high-resolution spatial data.

**Earlier applied projects** 2016 – 2020

- Panel-threshold evidence on green innovation and $CO_2$ (*TFSC* 2019, 1,000+ citations); synthetic-control estimate of the Wenchuan earthquake's long-run output effect; GMM analysis of trade-war and COVID farm-aid payments (MFP/CFAP) and U.S. agricultural lending; threshold-VAR/VECM/GARCH commodity-price models; equity-premium prediction from *NYT*/*WSJ* text.

## PUBLICATIONS

- *Biomass and Bioenergy* (2026), first author — spatial optimization of aviation-fuel supply chains
- *Transportation Research Record* (2024), first author — aviation-fuel conversion technologies
- *J. Agricultural and Applied Economics Assoc.* (2023) — beef price spreads and processing capacity
- *Technological Forecasting and Social Change* (2019) — green innovation and $CO_2$ (1,000+ citations)

## TEACHING

**Instructor:** ECON 351 Monetary Economics (Spring 2025, 2026); ECON 322 The Global Economy: Trade and Development (Summer 2025); ECON 313 Intermediate Macroeconomics (Summer 2026).
**Teaching assistant:** Ph.D. Microeconomic Theory and Mathematical Methods (ECON 511, 512, 581); Agribusiness Operations Research (AREC 525).

## HONORS

Charles B. Garrison Award for Excellence in Teaching (best teaching), University of Tennessee 2026
Summer Graduate Assistantship, University of Tennessee 2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee 2024
J. Fred and Wilma Holly Fellowship, University of Tennessee 2023 – present